

EduVision DV

Higher Education Performance Dashboard

Final Project Documentation

Covers data sources, data model, KPI methodology, all four analytical dashboards, testing and quality assurance, and project delivery.

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1. Executive Summary:

EduVision DV is a higher education analytics project built around a single unified Tableau workbook containing five dashboards: a home dashboard (EduVision DV) plus four analytical dashboards (University Overview, Research Analytics, Student Analytics, and Country Comparison). The workbook is fed by a combined dataset assembled from four public sources: **QS World University Rankings 2025**, **Times Higher Education (THE) World University Rankings**, **World University Rankings 2023**, and **World Bank Education Statistics (EdStats)**.

This document is the consolidated, final record of the project. *It replaces the separate milestone notes produced earlier and folds them into one reference:* what the data is and how it was cleaned, what each KPI means and how it is calculated, how each dashboard is built and how to use it, what was tested and what the results were, and how the finished workbook is organized and delivered.

1.1 Headline Results:

- Four public source datasets profiled, cleaned, and merged into one combined analytical dataset.
- 1,503 universities represented in the university dimension used across the four dashboards.
- 868 Research Analytics records and 842 Student Analytics records matched back to the university dimension with no fabricated identifiers.
- 216 countries represented in the unified country dimension, feeding 55,918 country-year-indicator observations in Country Comparison.
- Six core KPIs engineered and validated: Global Ranking Score, Research Impact Score, Faculty-to-Student Ratio, International Student Percentage, Academic Reputation Score, and Research Productivity Index.
- Five dashboards built and integrated behind a single set of global filters, a shared parameter driven navigation control, and consistent dark theme formatting.
- A structural defect in the home dashboard, overlapping floating chart zones causing a double exposure visual glitch, was identified during testing and corrected. Details are in Section 9.

1.2 Document Scope:

This document addresses every task listed under Milestone 4 (Testing and Delivery) of the project plan, and consolidates the outputs of Milestones 1 through 3 so a new reader does not need to open the earlier milestone files to understand the finished product. It is written to be portfolio ready: a reviewer with no prior context should be able to read this document, open the accompanying Tableau workbook, and understand exactly what they are looking at.

2. Project Overview:

2.1 Objective:

The project develops a comprehensive higher education analytics dashboard suite for analysing university rankings, research performance, student diversity, academic excellence, and global education trends, using publicly available ranking and education statistics datasets. The final deliverable is a single, unified Tableau workbook.

2.2 Audience:

The dashboard suite is designed to be usable by five audiences with different goals:

- Students: comparing universities before making an application decision.
- Academic researchers: benchmarking research output and citation impact across institutions.
- University administrators: tracking their own institution's position and trend over time.
- Policymakers: comparing country level education indicators for planning purposes.
- Education consultants: producing quick, defensible comparisons for clients.

2.3 The Five Dashboards:

#	Dashboard	Role	Primary Grain
1	EduVision DV	Home / landing page with global filters and navigation	N/A, control layer
2	University Overview	Global ranking, KPI summary, distribution	One row per university
3	Research Analytics	Teaching, research, citations, industry income	One row per university (THE)
4	Student Analytics	Enrollment, diversity, faculty ratio	One row per university (WUR 2023)
5	Country Comparison	Country level education indicators over time	One row per country/year/indicator

2.4 Project Timeline:

The project was organized into four milestones, each spanning two working weeks:

1. Milestone 1, Weeks 1 to 2: Data Collection and Preparation.
2. Milestone 2, Weeks 3 to 4: KPI Engineering and Dashboard Planning.
3. Milestone 3, Weeks 5 to 6: Dashboard Development.
4. Milestone 4, Weeks 7 to 8: Testing and Delivery (this document's primary subject).

3. Data Sources and Data Model

3.1 Source Datasets

Source	Provider	Approx. Scope	Used By
QS World University Rankings 2025	Quacquarelli Symonds	1,500+ institutions, global ranking and reputation scores	University Overview
THE World University Rankings	Times Higher Education	800+ institutions, teaching, research, citations, industry income	Research Analytics
World University Rankings 2023	Aggregated ranking release	800+ institutions, student body and diversity metrics	Student Analytics
World Bank Education Statistics (EdStats)	World Bank	200+ countries, multi year education indicators	Country Comparison

3.2 Combined Data Model:

The four source datasets are merged into one combined Tableau data source, EduVision_combined_dataset, so that a single Tableau workbook connection drives all five dashboards. Internally, the combined dataset preserves four logical tables joined through shared identifiers:

Table	Grain / Key Fields	Role
01 University Overview	university_id, country_id, region, year, rank_2024_numeric, rank_2025_numeric, overall_score	Central university level ranking table, hub of the model
02 Research Analytics	university_id, scores_research, scores_citations, scores_industry_income, rank_2024/2025_numeric	Research performance scores per institution
03 Student Analytics	university_id, total_students, student_staff_ratio, international_students_percentage, female_pct, male_pct	Student body and diversity indicators per institution
04 Country Comparison	country_id, country_name, region, income_group, year, indicator_code, indicator_name, value	Long format country level education and economic indicators

3.3 Relationship Pattern:

- 01 University Overview is the hub of the model.
- 02 Research Analytics and 03 Student Analytics relate to the hub many to many, matched on university_id and cross checked on country_id.
- 04 Country Comparison relates to the hub many to one on country_id. It has no institution level rank field; it stores indicator and value pairs (13 confirmed indicators, for example Gross enrolment ratio tertiary both

sexes percent, Government expenditure on education as percent of GDP, and Adult literacy rate population 15 plus years both sexes percent) across 7 World Bank regions and 104 countries.

3.4 Data Cleaning and Transformation:

Before the combined dataset was built, each source went through a standard cleaning pipeline:

1. Remove duplicate institution and country records.
2. Standardize university names so the same institution is not represented under two different labels across sources.
3. Standardize country names against a single reference list.
4. Normalize ranking and score metrics to comparable numeric ranges.
5. Resolve name based matching ambiguity using a combination of country, region, and fuzzy name matching, with manual review of any low confidence match.

The cleaning pass reduced missing values to under 2 percent across key fields, meeting the target set in Milestone 1.

3.5 Known Data Quality Issues and Resolutions:

Issue	Description	Resolution
Cross country false match	One university was matched to an institution of the same name in a different country during automated matching.	Corrected manually after review, university reassigned to the correct country_id.
Character encoding corruption	Special characters in some university and country names (accented letters) were corrupted on import.	Re-imported affected columns with explicit UTF-8 encoding and re-validated against source text.

3.6 Referential Integrity Result:

The final combined dataset passed referential integrity validation: 0 invalid university identifiers, 0 invalid country identifiers, and 0 unresolved name mismatches across the four source tables.

4. KPI Definitions and Methodology:

Six KPIs were engineered from the cleaned source data during Milestone 2 and are used consistently across the four analytical dashboards. Each KPI below lists its definition, the fields it is derived from, and where it appears in the workbook.

4.1 Global Ranking Score:

Definition: the institution's overall position in the combined ranking, built from the QS 2025 and THE ranking positions. Displayed as both a numeric rank and an overall score out of 100.

Derived from: rank_2024_numeric, rank_2025_numeric, overall_score.

Appears in: University Overview (Top Global Ranking, Top University Rankings 2024/2025).

4.2 Research Impact Score:

Definition: a composite measure of an institution's research output and influence, combining research volume, citation strength, and industry income.

Derived from: scores_research, scores_citations, scores_industry_income.

Appears in: Research Analytics (Avg. Research Score, Avg. Citations Score, Research Impact Comparison).

4.3 Faculty-to-Student Ratio:

Definition: the number of students per faculty member at an institution, expressed as a ratio such as 1 to 17.3. A lower number of students per faculty member generally indicates more individual attention.

Derived from: student_staff_ratio, total_students.

Appears in: Student Analytics (Faculty-to-Student Ratio Analysis), and as a summary KPI card on University Overview (Avg. Faculty Student Ratio).

4.4 International Student Percentage:

Definition: the share of an institution's total enrolled students who are international students.

Derived from: international_students_percentage, total_students.

Appears in: Student Analytics (International Student Analysis) and as a KPI card on University Overview (Total International Students Percentage).

4.5 Academic Reputation Score:

Definition: a survey based measure of how highly an institution is regarded by academic peers, one of the core inputs to the QS ranking methodology.

Derived from: academic_reputation, employer_reputation.

Appears in: University Overview (Academic Reputation Analysis).

4.6 Research Productivity Index:

Definition: a normalized measure of research output volume relative to institution size, allowing fair comparison between large and small institutions.

Derived from: scores_research combined with total_students.

Appears in: Research Analytics (Research Productivity Trends, Publications Analysis).

4.7 KPI Engineering Process:

KPIs were calculated in Python (pandas and numpy) prior to loading into Tableau, rather than as Tableau calculated fields. This keeps the Tableau workbook fast and keeps the KPI logic version controlled and testable outside of Tableau. The KPI generation script (generate_education_kpis.py) writes the final, Tableau ready dataset (university_final_dataset.xlsx), which is the single file the combined Tableau data source connects to.

5. Dashboard Architecture:

5.1 Navigation Model:

The workbook opens on the EduVision DV home dashboard. From there, the four analytical dashboards are reached either through the bottom dashboard tab strip, or through the DASHBOARDS parameter control and Navigation buttons placed on the home page. A shared Top N parameter and a Select Education Indicator parameter appear on the home page and are referenced by charts on the analytical dashboards.

5.2 Global Filters and Parameters:

Control	Type	Purpose
Year	Quick filter	Restrict all dashboards to a single reporting year (default 2024).
Region	Quick filter	Restrict to one of the six world regions, or All.
Country	Quick filter	Restrict to a single country, or All.
Subject Area	Quick filter	Restrict Research Analytics content to one subject area, or All.
Top N	Parameter, slider	Controls how many top ranked universities are shown in leaderboard style charts.
Select Education Indicator	Parameter, dropdown	Chooses which World Bank indicator is charted in Country Comparison.
DASHBOARDS	Parameter, dropdown	Displays the name of the currently intended destination dashboard from the home page menu.

5.3 Visual Design Standard:

All five dashboards share a single formatting standard, set once at the workbook level rather than sheet by sheet: dark navy background, white body text, purple accent color for titles and highlights, and no visible gridlines or row dividers. This was applied through Format, Workbook in Tableau so that every worksheet inherits the same look automatically.

6. Dashboard Guide: University Overview

University Overview is the primary landing point for ranking based questions: who is ranked where, how the global ranking is distributed geographically, and how individual institutions compare on reputation and performance trend.

6.1 Worksheets on This Dashboard:

Worksheet	Chart Type	What It Shows
Top Global Ranking	Single value card	The number 1 ranked university for the selected year, with name and country.
Total Universities	Single value card	Count of ranked universities in the current filter context.
Average Overall Score	Single value card	Mean overall score across all ranked universities, with year over year change.
Total International Students Percentage	Single value card	Average international student share across all institutions.
Avg. Faculty Student Ratio	Single value card	Average faculty to student ratio across all institutions.
Top University Rankings in 2024	Ranked table with bar	Top N universities by 2024 rank, with country flag and overall score bar.
Top University Rankings in 2025	Ranked table with bar	Same view for the 2025 ranking cycle, for year over year comparison.
Global University Distribution	Filled map / donut	Count and percentage share of universities by world region.
Academic Reputation Analysis	Horizontal bar	Top institutions ranked by academic reputation score.
University Performance Trends	Line chart	Overall score trend by year for a small set of highlighted universities.
Institutional Comparison	Bar chart	Side by side comparison of publications by the top five universities.

6.2 Usage Notes:

- Use the Year filter to compare a single year at a time, the KPI cards and Top University Rankings tables both respond to it.
- Clicking a university row in either ranking table highlights that same university across the other charts on the dashboard via a dashboard filter action.
- The Top N parameter, set on the home page, controls how many rows appear in both ranking tables.

7. Dashboard Guide: Research Analytics

Research Analytics answers questions about research output: which institutions publish the most, which are cited most, and how research strength is distributed by world region.

7.1 Worksheets on This Dashboard:

Worksheet	Chart Type	What It Shows
Publications Analysis	Bar chart	Average research score by university, sorted descending.
Citation Performance	Scatter plot	Average research score against citation score, coloured by region, one dot per university.
Research Productivity Trends	Grouped bar chart	Citation score by university across the current filter selection.
Top Research Institutions	Ranked table with heat coloring	Research score, citation score, and industry income score side by side per institution.
Research Impact Comparison	Treemap	Average citation score aggregated by world region, sized by region.
Avg. Research Score / Avg. Citations Score / Avg. Industry Income	Single value cards	Headline averages for the current filter context, shown above the chart grid.

7.2 Usage Notes:

- The scatter plot (Citation Performance) is the best single chart for spotting institutions that are strong in research but weak in citations, or the reverse.
- The Region legend is a highlight control, click a region to isolate it across every chart on this dashboard.
- This dashboard shares its Year, Region, Country, and Subject Area filters with the other three analytical dashboards.

8. Dashboard Guide: Student Analytics

Student Analytics answers questions about the student body: how international an institution is, how it compares on faculty availability, and how the student population is distributed by gender and region.

8.1 Worksheets on This Dashboard:

Worksheet	Chart Type	What It Shows
International Student Analysis	Map or bar chart	International student percentage by region or by institution.
Faculty-To-Student Ratio Analysis	Bar chart	Faculty to student ratio by region, expressed as 1 to N.
Student Diversity Trends	Line or area chart	Female and male percentage trend over time.
Enrollment Comparisons	Bar chart	Total students by institution or region.
Student Distribution Analysis	Donut or bar chart	Share of total global student population by region.
Avg. Female Percentage / Avg. Male Percentage	Single value cards	Headline gender split averages for the current filter context.
Total Students / Total Global Students	Single value cards	Aggregate enrollment counts.

8.2 Usage Notes:

- Faculty to Student Ratio is displayed as a score (for example 17.3) rather than a ratio, to match how the metric is normally read in higher education reporting.
- The gender percentage fields (Female Pct, Male Pct) are sourced from the World University Rankings 2023 dataset and may not be reported for every institution, unreported institutions are excluded from the diversity charts rather than shown as zero.

9. Dashboard Guide: Country Comparison

Country Comparison answers questions at the national policy level: how countries compare on a chosen education indicator, and how that indicator has moved over time.

9.1 Worksheets on This Dashboard:

Worksheet	Chart Type	What It Shows
Country Ranking Comparison	Bar chart	Countries ranked by the currently selected education indicator.
Education Performance Benchmarking	Line chart	Selected indicator trend over time for a small set of highlighted countries.
Regional Education Trends	Area or line chart	Selected indicator aggregated by world region over time.
Top Performing Countries	Ranked table	Leaderboard of countries on the selected indicator.
Total Countries Covered	Single value card	Count of distinct countries present in the current filter context.
Selected Indicator Average / Maximum Indicator Value / Minimum Indicator Value	Single value cards	Summary statistics for whichever indicator is chosen in the Select Education Indicator parameter.
High Income by OECD / High Income by nonOECD	Bar or comparison chart	Splits high income countries by OECD membership for the selected indicator.

9.2 Usage Notes

- This is the only dashboard driven by the Select Education Indicator parameter, changing it on the home page changes every chart on this dashboard, since all of them read the same [indicator_name] = [Parameter] calculated field.
- Because Country Comparison uses a long format table (13 indicators stacked as rows rather than columns), always confirm the correct indicator is selected before reading values off this dashboard.

10. Filters, Parameters, and Dashboard Actions:

10.1 Filter Actions:

Several dashboard actions were configured to make the workbook feel like one connected application rather than four separate charts:

- Selecting a university on a ranking table filter or highlights related charts on the same dashboard (Filter and Highlight actions).
- Selecting a region on any region-coloured chart highlights that region across the other charts on the same dashboard.
- Selecting a country-on-Country Comparison filters the related university level charts where a country mapping exists.

10.2 Parameter Actions:

The Select Education Indicator and Top N parameters are exposed as controls on the home page rather than repeated on every dashboard, keeping the analytical dashboards visually focused on their charts.

10.3 Design Rationale:

Filters were kept at exactly four (Year, Region, Country, Subject Area) by deliberate choice. Additional filters were considered during Milestone 2 planning but were dropped to avoid overwhelming a first-time user, four filters cover the overwhelming majority of comparison questions the intended audiences ask.

11. Testing and Quality Assurance Summary:

Full detail for this section lives in two companion documents delivered alongside this one: the QA Checklist and the Dashboard Testing Report. This section summarizes their outcome.

11.1 What Was Tested:

- KPI calculations, spot checked against the source spreadsheet for a random sample of institutions and countries.
- Ranking calculations, confirming rank_2024_numeric and rank_2025_numeric sort correctly and match the source ranking order.
- Dashboard interactions: every filter, parameter, and action described in Section 10.
- Educational metrics: faculty to student ratio, international student percentage, and the six core KPIs, checked for plausible ranges (for example, no negative percentages, no ratios below 1 to 1).

11.2 Defect Found and Corrected:

During interaction testing, the EduVision DV home dashboard was found to render with visible overlap: multiple charts from different analytical dashboards appeared stacked on top of one another when the DASHBOARDS parameter changed value. Root cause analysis traced this to floating layout zones placed at fixed coordinates with no dashboard action or Tableau feature actually toggling their visibility, so all of them rendered simultaneously regardless of the selected parameter value.

Resolution: the home dashboard's zone layout was rebuilt so it holds only the header, the three shared controls (Top N, Select Education Indicator, DASHBOARDS), and the region selector, with every duplicate floating chart zone removed. Page to page navigation now uses the existing dashboard tab strip, with Tableau Navigation button objects recommended as the next enhancement (see Section 13.3).

11.3 Result:

Metric	Target	Actual
KPI accuracy	Above 95 percent	100 percent on sampled checks, 0 discrepancies found
Dashboard functionality	4 dashboards integrated	5 dashboards integrated (4 analytical plus home)
Major outstanding defects	None	None open, 1 found and resolved during testing

Full test case detail, including test IDs, steps, expected results, and pass/fail status for every dashboard, is recorded in the Dashboard Testing Report.

12. Known Issues and Cleanup Items:

The workbook file currently contains a small number of leftover working tabs from earlier development iterations. These should be removed before the workbook is published or shared externally, since they are not part of the intended final deliverable and could confuse a first-time viewer:

Tab Name	Type	Recommended Action
COPY dont del	Dashboard, draft copy	Delete before final publish
Dashboard dont del	Dashboard, draft copy	Delete before final publish
INFO	Dashboard, working notes	Delete or fold content into this documentation
UNIVERSITY OVERVIEW / RESEARCH ANALYTICS / STUDENT ANALYTICS / COUNTRY COMPARISON (uppercase)	Dashboard, duplicate of the title case versions	Delete duplicates, keep only the title case versions listed in Section 5

“None of these items affect the correctness of the data or the four core dashboards”, they are housekeeping items only. Not removing them is recommended because from these we can track them. As the last step before the GitHub and Tableau Public deployment described in Section 13.

13. Project Structure and Delivery:

13.1 Repository Structure:

The project repository is organized into four top level folders:

Folder (may Vary)	Contents
/scripts	data_collection.py, education_cleaning.ipynb, generate_education_kpis.py
/data	university_raw_data.csv, university_cleaned.csv, university_final_dataset.xlsx
/dashboard	EduVision_DV.twbx (final packaged Tableau workbook)
/docs	This Final Documentation, the QA Checklist, the Dashboard Testing Report, and dashboard_storyboard.pdf

13.2 GitHub Deployment:

The repository should be pushed to GitHub with a top level README.md summarizing the project (a suggested README is delivered alongside this document), and with the four folders above committed as shown. The .twbx file is a binary archive and should be tracked normally, no special Git LFS configuration is required at the current file size.

13.3 Tableau Public Deployment (Optional):

The workbook is published to Tableau Public, complete the cleanup in Section 12 first, then add Navigation button objects to the EduVision DV home dashboard so that a Tableau Public viewer, who does not see the desktop tab strip prominently, has an obvious in canvas way to move between dashboards.

Link to Access:

https://public.tableau.com/views/UniversityOverview_17890783104670/UNIVERSITYOVERVIEW?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

13.4 Final Deliverables Checklist:

Deliverable	Status
university_raw_data.csv, data_collection.py	Delivered, Milestone 01
university_cleaned.csv, education_cleaning.ipynb	Delivered, Milestone 01
university_final_dataset.xlsx, generate_education_kpis.py	Delivered, Milestone 02
dashboard_storyboard.pdf, eduvision_prototype.twbx	Delivered, Milestone 02
eduvision_dashboard_v1.twbx	Delivered, Milestone 03

Deliverable	Status
EduVision_DV.twbx (integrated, all dashboards linked)	Delivered, Milestone 03, revised Milestone 04
QA Checklist	Delivered, Milestone 04
Dashboard Testing Report	Delivered, Milestone 04
GitHub Repository (structure and README)	Delivered, Milestone 04
Final Documentation (this document)	Delivered, Milestone 04

14. Evaluation Criteria and Milestone Summary:

Milestone	Focus Area	Metric	Target	Result
1	Data Collection and Cleaning	Dataset completeness	Above 95 percent complete	Met
2	KPI Engineering	KPI accuracy	6 or more KPIs generated	Met, 6 KPIs delivered
3	Dashboard Development	Dashboard functionality	4 dashboards integrated	Met, 5 dashboards integrated including home
4	Documentation and Delivery	Project quality	Portfolio ready	Met, see this document

14.1 Tech Stack:

Area	Tools / Libraries
Data collection	Python, QS Rankings, World University Rankings, World Bank EdStats
Data processing	Pandas, NumPy
Data cleaning	Python
Visualization	Tableau Public
Dashboard integration	Tableau filters, parameters, and dashboard actions
Documentation	Markdown, Microsoft Word, GitHub

15. Appendix A: Full Worksheet Inventory

For reference, every worksheet present in the final workbook, grouped by the dashboard it appears on.

A.1 EduVision DV (Home):

- REGION (region selector icons)

A.2 University Overview:

- Top Global Ranking, Total Universities, Average Overall Score, Total International Students Percentage, Avg. Faculty Student Ratio
- Top University Rankings in 2024, Top University Rankings in 2025
- Global University Distribution, Academic Reputation Analysis
- University Performance Trends, Institutional Comparison

A.3 Research Analytics:

- Publications Analysis, Citation Performance, Research Productivity Trends
- Top Research Institutions, Research Impact Comparison
- Avg. Research Score, Avg. Citations Score, Avg. Industry Income, Avg. Teaching Score, Avg. International Outlook, Avg. International Outlook 03

A.4 Student Analytics:

- International Student Analysis, Faculty-To-Student Ratio Analysis
- Student Diversity Trends, Enrollment Comparisons, Student Distribution Analysis
- Avg. Female Percentage, Avg. Male Percentage, Avg. Employment Outcome, Total Students, Total Global Students

A.5 Country Comparison:

- Country Ranking Comparison, Education Performance Benchmarking, Regional Education Trends
- Top Performing Countries, Total Countries Covered
- Selected Indicator Average, Maximum Indicator Value, Minimum Indicator Value
- High Income by OECD, High Income by nonOECD, Data Coverage Timeline

16. Appendix B: Glossary

Term	Meaning
Grain	The level of detail one row in a table represents, for example one row per university.
Dimension	A descriptive field used to group or filter data, such as country or region.
Measure	A numeric field that can be aggregated, such as overall score or total students.
Parameter	A workbook wide input control whose value can be referenced by calculated fields across multiple sheets.
Dashboard action	A rule that makes selecting a mark on one chart filter, highlight, or navigate another chart or dashboard.
Referential integrity	The property that every identifier used to join two tables actually exists in both tables, with no orphaned or fabricated values.

17. Version History:

Version	Date	Summary of Changes
1.0	Milestone 1	Initial data collection and cleaning documentation.
1.5	Milestone 3	Added data model reference and per dashboard worksheet configuration.
2.0	Milestone 4	Consolidated all prior milestones into one final document, added KPI methodology, full dashboard guide for all four analytical dashboards, testing and QA summary, known issues, repository structure, and evaluation criteria.

“End of document”